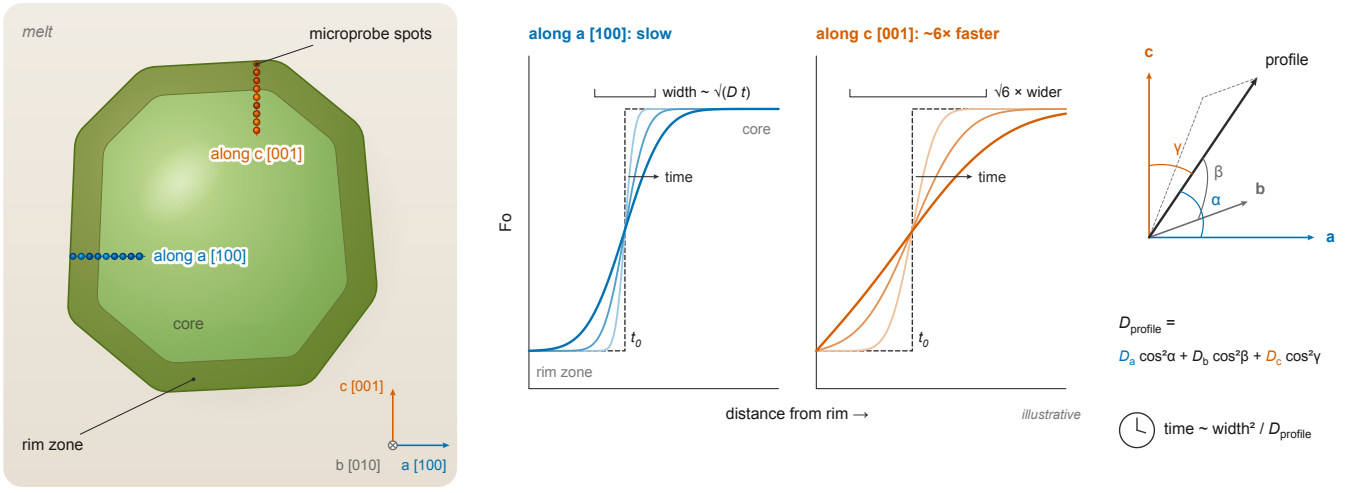

a Zoned olivine in melt
melt
microprobe spots
along c [001]
along a [100]
core
rim zone
c [001]
b [010]
a [100]
b Step relaxes with time
along a [100]: slow
along c [001]: ~6× faster
width ~ √(D t)
√6 × wider
core
time
time
Fo
t_0
t_0
rim zone
distance from rim →
illustrative
c Direction sets D
c
profile
γ
β
α
b
a
$D_{profile}$ =
$D_a \cos^2\alpha + D_b \cos^2\beta + D_c \cos^2\gamma$
time ~ width² / $D_{profile}$